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MID SEMESTER EXAMINATION Sept-2025

Name of The Program : B.Tech (Civil. Engg)
Name OF the Course : Reinforced Cement Concrete Design
Time: 90 Min

Semester : V
Course Code : TCE 502
Max Marks : 50

- Note: i) Answer all the questions by choosing any one of the sub questions.
ii) Each question carries 10 Marks.
iii) IS code 456-2000 supplied by the department is permitted for design use.

Q1 Answer any one. (1X10=10 Marks)
(a) Give an elaborate discussion of various field tests and equipment's that shall be applied in RCC construction site? (co1)

(OR)

(b) What is effect of Size and Shape of the concrete specimen in the determination of the characteristic strength of concrete in the laboratory? (co1)

Q2 Answer any one. (1X10=10 Marks)
(a) Differentiate the Balanced, Under and Over Reinforced beam sections of a Beam? (co1)

(OR)

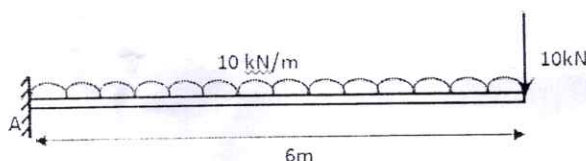
(b) Explain how the Idealized Stress-Strain curve for the Steel is developed in IS 456-2000? (co1)

Q3 Answer any one. (1X10=10 Marks)
(a) Derive the design constants for a singly reinforced balanced rectangular beam section of $b \times D$ dimensions using either in Working Stress method or Limit State Method of RCC Design philosophy? (co2)

(OR)

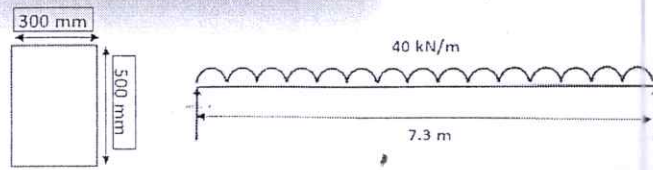
(b) Explain the concept of "Curtailement and Bent up Main Reinforcement" in RCC Beam Design? Explain in Detail about the design criteria's and provision for the shear reinforcement for various methods of reinforcement detailing for beam Shear? (co2)

Q4 Answer any one. (1X10=10 Marks)
(a) Design of a following beam section for maximum bending moment, assuming M20 and Fe415 Grade Steel materials for flexure? (co2)



(OR)

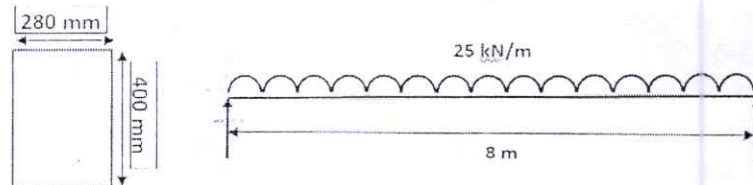
- (b) A simply supported beam has the following section details. It is provided with 5-32 mm dia main bars (with 2 bent up at ends) and 4-20 mm dia compression reinforcements. Please check and design the shear reinforcements for the section if necessary? (co2)



Q5 Answer any one.

(1X10=10 Marks)

- (a) Using M25 grade Steel and Fe415 Grade Steels, Design the beam for flexure with the given dimensional ? (co2)



(OR)

- (b) Decide the Idealized UDL for the Beam AB in the given building plan for the following slab Load Data. The slab thickness is taken as 150mm? (co2)

- I) Live Load on Slab – 54 kN/m²
- II) Dead Load on Slab – 2 kN/m²
- III) Floor Finish Load on Slab – 1.5 kN/m²
- IV) Self weight of Slab

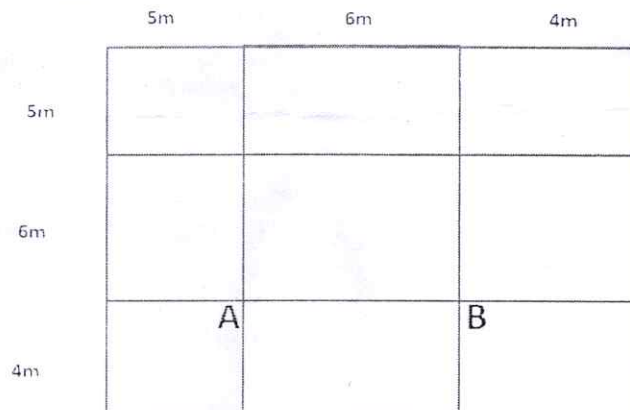


Table: For general LSM Design Use

Stress-Strain Values for $F_y = 500$					Stress-Strain Values for $F_y = 415$				
K	Strain	Ultimate Stress	Idealized Curve	Corresponding Strain	K	Strain	Ultimate Stress	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000	0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	400.0000	347.83	0.001839	0.8000	0.0001	332.0000	288.70	0.001543
0.8500	0.0003	425.0000	369.57	0.002148	0.8500	0.0003	352.7500	306.74	0.001834
0.9000	0.0007	450.0000	391.30	0.002657	0.9000	0.0007	373.5000	324.78	0.002324
0.9500	0.0010	475.0000	411.04	0.003065	0.9500	0.0010	394.2500	342.83	0.002714
0.9750	0.0020	487.5000	423.91	0.004120	0.9750	0.0020	404.6250	351.85	0.003759
1.0000	0.0035	500.0000	434.78	0.005674	1.0000	0.0035	415.0000	360.87	0.005304